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The polarization of energy preferences – A study on social acceptance of wind and nuclear power in Sweden

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ABSTRACT

Using Sweden as a study case, this article explores the polarized opinions to wind and nuclear energy, two low carbon energy options that have been shown to be politically controversial. In a wide-scale survey (N = 5200), general attitudes to wind and nuclear energy are captured, as well as to projects in the proximity of people's homes. The study demonstrates a deep polarization of energy preferences in Sweden, finding strong associations between worldviews, political orientation, environmental concern, and support for or resistance to wind and nuclear energy. The study concludes that support for both energy options is reduced when wind or nuclear power is constructed near people's home, but also suggests that the proximity effect is particularly strong for individuals with strong TAN (traditional, authoritarian, nationalistic) values and right leaning political ideology. The article argues that politically motivated reasoning might explain the polarization of attitudes, yet this effect seems to become less relevant when people are asked to judge potential energy infrastructure located close them.

1. Introduction

With the increasing urgency of a transition away from fossil fuels, the discussion of various options of low-carbon energy technologies has attracted the interest of the wider society. Public engagement in energy policy debates is motivated by climate concerns, but also by rising energy prices and the relevance of energy independence, especially following the Russian invasion of Ukraine (Osička and Černoch, 2022). The energy preferences in society are, however, not only guided by economic and environmental dimensions, but also by different beliefs and values. Investments and expansions of various energy technologies can therefore provoke different socio-psychological responses (Dermont et al., 2017; Batel, 2020), and be subject to political polarization (Hawes and Nowlin, 2022). Energy systems are accordingly not just physical infrastructures, but socio-technical systems, entangled with economic relations, power structures, social practices, norms, and institutions (Parag and Sovacool, 2016; Newell et al., 2022).

The social dimensions of energy systems have attracted the interest of researchers, with a particular focus on the concept of social acceptance of different energy technologies (Batel et al., 2013; Fast, 2013). Renewable energy, such as wind power, have become increasingly cost-effective (Holechek et al., 2022), and at COP28, more than 110 countries worldwide pledged to triple the world's renewable energy capacity by 2030 (Abnett et al., 2023). However, investments in and development of renewable energy, particularly wind power, are often stalled due to resistance from local communities. Wind power has proven to be controversial, triggering disputes and opposition among groups living near proposed projects (Krekel and Zerrahn, 2017). These groups may be negatively disposed due to auditory or visual disturbances (Carley et al., 2020), and concerned about impacts on biodiversity, wildlife, and land use (Dugstad et al., 2020) or place-attachment (Devine-Wright, 2009; Dugestad et al., 2023).

At the same time, negative attitudes toward wind power have been shown to elicit value-related emotions, associated with perceptions of distributional or procedural injustice (Wüstenhagen et al., 2007; Knauf, 2022), ideological leanings and trust (Clulow et al., 2021; Vuichard et al., 2019), or environmental concerns (Azarova et al., 2019). Resistance to wind power has also been incorporated into political campaigns of right-wing populist parties and has become part of a broader skepticism toward the climate and energy transition (Batel and Devine-Wright, 2018; Lockwood and Lockwood, 2022).

Another low-carbon energy source forecasted to play a role in the

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energy transition is nuclear energy (Buongiorno et al., 2019; Murshed et al., 2022). Due to high costs, risks, and lengthy construction processes, nuclear energy has been in decline for several years, yet it could become increasingly relevant with the development of new technologies, such as small modular reactors (IAE, 2023). At COP28 in Dubai, 22 countries signed a pledge to triple nuclear energy capacity by 2050, recognizing the role of nuclear power in keeping the 1.5 °C goal alive (NEA, 2023). The use of nuclear power is, however, disputed due to the risk of accidents and unresolved waste management issues (Gupta et al., 2019; Wang and Kim, 2018). Interestingly though, attitudes toward nuclear power appear to be determined more by political and ideological factors, values, and worldviews than by risk perceptions and economic considerations (Whitfield et al., 2009). Overall, the support for further nuclear investments in the European Union is fairly weak (Eurobarometer, 2022).

To successfully navigate a pathway toward a low-carbon and sustainable future, it is essential to understand how different social and value-based dimensions influence energy policies. This article aims to contribute to the literature by exploring factors that determine energy opinions, focusing on the polarization of wind and nuclear preferences in Sweden. Sweden is an interesting case study, as it is a country with a high level of both wind and nuclear energy in its energy mix, yet one where energy policies are marked by deep political tensions and polarization. The study draws conclusions from a large-scale survey conducted in Sweden during 2023, including over 5200 respondents, who were randomly selected and representative of the wider Swedish population.

The results show that low-carbon energy investments in Sweden are likely to encounter resistance due to a sizable antagonistic minority who are strongly opposed to either wind or nuclear energy. Interestingly, among those with traditional, nationalistic, and authoritarian values and right leaning political ideology, the enthusiasm for nuclear energy seems to reduce the closer a new nuclear power plant would be to their own residences. The study highlights the importance of recognizing the sociopsychological dimensions within political frameworks aiming for a transition toward a low-carbon energy system.

2. Overview of relevant literature

Attitudes to wind and nuclear power are divisive in many countries. Both of these energy technologies generate different social emotions in different societal groups, often related to beliefs about benefits and risks associated with the respective technology. Factors influencing attitudes to different energy options tend to be contextual and they vary across different countries (Arndt, 2023; Wang and Kim, 2018; Nguyen and Yim, 2018).

Opinions about nuclear energy are particularly influenced by normative aspects related to the risk of accidents, uranium extraction, the treatment of radioactive waste, and its association with the military industry (Siegrist and Visschers, 2013; Keller et al., 2012; Groh and Möllendorff, 2020; Siegrist et al., 2014). Support for nuclear energy has clearly been affected by the three major accidents worldwide—Three Mile Island, Chernobyl, and Fukushima—after which public attitudes turned more negative (Richard Eiser et al., 1990; Visschers and Siegrist, 2013). Nuclear power can, on the other hand, also be associated with energy security and economic prosperity (Slovic et al., 2000; Arndt, 2023), and social acceptance can, thus, be described as a trade-off between risks and potential economic benefits (Finucane et al., 2000; Slovic et al., 2004).

People's perceptions of risks and benefits differ, however, depending on individual-level factors. Aspects that have been demonstrated to influence opinions are political ideology, gender, income, and education (Costa-Font et al., 2008; Sundström and McCright, 2016; Gupta et al., 2019; Spence et al., 2010; Hazboun and Boudet, 2020), as well as institutional and country-level factors, such as the national energy mix and political aspects (Wang and Kim, 2018; Bohdanowicz et al., 2023).

Renewable energy is not associated with similar risk perceptions, and in many countries, wind energy is among the most popular energy options (Arndt, 2023). Attitudes expressed at nation-level tend to diverge, however, from the views of the communities living in the proximity of wind power investment (Krekel and Zerrahn, 2017; Dugstad et al., 2020). Bell et al. (2005) expressed accordingly that a "social gap" tends to arise between the high level of public support at the national level, and the low level of successful implementation of wind energy projects.

The distance of wind energy sitings from peoples' residence, the so-called proximity effect, has shown to have a strong impact on opinions (Carley et al., 2020; Karasmanaki and Tsantopoulos, 2021). Early studies on social acceptance of wind power focused accordingly on the proximity effect or so-called NIMBY-ism (not-in-my-back-yard) (Batel et al., 2013; Fast, 2013). However, there is criticism of the "NIMBY--term", which is sometimes used as a method to discredit opponents (Horst, 2007). While noise and visual intrusion of wind farms have shown to influence public attitudes (Carley et al., 2020; Krekel and Zerrahn, 2017), opposing responses are often based on more complex aspects (Dermont et al., 2017; Batel, 2020). Devine-Wright (2009), Horst (2007) and Dugstad et al. (2023) show that socio-cultural attachment to the place of living is of relevance, and that opposition can be seen as "place-protective action". Public acceptance of wind power is often influenced by the perceived environmental impacts of the wind farm location, and to what extent biodiversity in landscapes are affected (Dugstad et al., 2020). Biodiversity concerns differ, however, between different communities, and in Sweden, they are of particular relevance for Sami communities (Ramasar et al., 2022; Ek and Matti, 2015). Hence, careful analysis of the spatial allocation of wind farms is important when addressing inherent trade-offs between local disamenities, biodiversity conservation, and climate change considerations (Kati et al., 2021; Grimsrud et al., 2024).

Geographic proximity does not have similar impacts on attitudes on nuclear energy, and studies show instead that people living close to nuclear power plants are more positively inclined than those living further away (Greenberg, 2009; Uji et al., 2021; Venables et al., 2012; Cale and Kromer, 2015). However, few studies have captured attitudes in relation to proximity prior to the construction process. It is likely that new nuclear power plant establishments in landscapes, just like new wind-power plants, may evoke negative reactions motived by sense of place or concerns about their impacts on local wild life and biodiversity (Sovacool, 2012).

Recent studies indicate, moreover, that opposition to wind power is associated with notions of being neglected or misrepresented (Walker and Devine-Wright, 2008). There are also research suggesting that perceptions of unfair distribution of the benefits and burdens of wind power influence acceptance (Huijts et al., 2012; De Luca et al., 2020). If profits from wind power generation is not redistributed locally, but transferred to the national level or to business corporations, individuals tend to see the construction of wind power as a kind of exploitation of the local landscape (Lindvall, 2023; Liebe et al., 2017; Knauf, 2022).

The decision-making process and the ability of affected individuals to participate in it, also impact acceptance of wind power (Walker and Baxter, 2017; Brennan and Rensburg, 2016; Le Maitre et al., 2023; Vuichard et al., 2022). Individuals who distrust public institutions, or who perceive the decision-making processes as unfair, non-transparent, or illegitimate, are also more likely to oppose wind farms (Gross, 2007; Liebe et al., 2017). Perceptions of procedural and distributional justice have also been shown to influence public attitudes to nuclear energy (Visschers and Siegrist, 2012; Besley, 2012), although the research on this matter is limited.

Attitudes to different energy options are, however, not only determined by aspects related to fairness and risks, but also by other types of personal norms, values, and ideological orientations. A particularly strong determinant for both wind and nuclear energy attitudes is environmental concern. Individuals who prioritize environmental protection

over economic aspects, are found to be more skeptical about nuclear energy, while those who perceive nuclear energy as economically beneficial are more prone to downplay environmental hazards (Pampel, 2011; Badora et al., 2021; Besley and Oh, 2014; Corner et al., 2011). Individuals with pro-environmental attitudes tend to, likewise, be more positive to wind energy (Azarova et al., 2019; Knauf, 2022), although concerns of the local environmental impacts may also motivate resistance (Dugstad et al., 2020). Some studies indicate that climate change concern has become less relevant for nuclear power acceptance (Bohdanowicz et al., 2023), yet other studies have not found evidence of this trend (Sonnberger et al., 2021; Arndt, 2023).

Several studies suggest, moreover, that attitudes toward nuclear energy are associated with political ideology. Individuals who define themselves politically to the left tend to be more positive to wind power (Karlstrøm and Ryghaug 2014; Clulow et al., 2021), while the opposite is true for nuclear energy (Whitfield et al., 2009; Costa-Font et al., 2008; Pampel, 2011; Clulow et al., 2021). Worldviews are also relevant, and individuals who express individualistic or hierarchical values tend to be more supportive of nuclear energy (Van De Graaff, 2016; Whitfield et al., 2009), while wind power is preferred by individuals who express egalitarian and ecological values (Bidwell, 2013).

As a result, attitudes to wind and nuclear energy are prone to political polarization (Perlaviciute and Steg, 2015; Roddis et al., 2019). The association between ideology and values, and public attitudes to different energy technologies, can therefore be influenced by political narratives or political cues (Latré et al., 2019), and media messaging and political debates (Ho and Kristiansen, 2019; Żuk, 2023; Kojo et al., 2020). Some studies have also shown that attitudes can be exploited by economic agents or interest groups advocating for or against the usage of renewable energy (Johnstone et al., 2017).

3. Energy policies in Sweden

3.1. The polarised debate on nuclear energy

Sweden is a country where energy policies have been fraught with difficult political disputes for over five decades. In the 1970s, Sweden developed a nuclear energy program, commissioning 12 nuclear reactors. These reactors were constructed between 1972 and 1980, making Sweden the country with the highest percentage of nuclear power usage per capita in the world (Kåberger, 2002). The construction of nuclear power plants was partly motivated by the needs and demands of the industry, but it was also a response to the 1973 oil price shock and the decision to halt the expansion of hydropower to protect the few remaining unexploited rivers in Sweden (Vedung, 2002). Initially, there was strong interparty consensus for the nuclear energy program; however, the expansion became an increasingly contentious political issue, partly spurred by the anti-nuclear stance taken by the Centre Party (Vedung, 1979). From 1976, nuclear energy emerged as one of the most disruptive issues in Swedish politics, causing a government crisis in 1978 and profoundly affecting the 1979 election results (Holmberg, 2022). Following the accident at Three Mile Island, anti-nuclear sentiments grew stronger, and to defuse the issue, a public referendum was held in 1980. A slight majority of voters supported a gradual phase-out of nuclear power in the referendum; however, the process of phasing out nuclear energy did not begin until 1999, when the first reactor was decommissioned. Since then, another five reactors have been closed down (Holmberg, 2009).

At the time of the referendum, opinions on nuclear energy were not initially influenced by ideological orientations but were determined by factors such as risk perception and environmental concern (Holmberg, 2018). Over time, the issue became increasingly politically polarized, and by 2021, right-leaning voters were, according to one survey, 58 percentage points more likely to favor nuclear power than left-leaning voters (Holmberg, 2022). In an attempt to depoliticize energy policies, an interparty energy agreement was negotiated by all major political parties in 2016. According to this agreement, the existing reactors were allowed to be replaced, but without any public subsidies. A long-term target of 100 percent renewable electricity production by 2040 was also adopted, and the so-called electricity certification system, benefiting renewable energy production, was prolonged and expanded (prop: 2017/18:228). In 2019, the agreement was dissolved following the departure of two conservative parties, which advocated for the continued operation of two reactors at the Ringhals power plant. The debate on nuclear energy then intensified, and in the 2022 election campaign, the pro-nuclear conservative parties ran on a proposal to construct new reactors, while the center-liberal, green, and left-leaning parties took neutral or skeptical stances (Holmberg, 2022). The debate was also influenced by the rising popularity of the Sweden Democrats, a nationalist and right-wing populist party, which strongly advocated for nuclear power and opposed wind power development. The conservative government that took power in 2022, with collaborative support from the Sweden Democrats, declared its intention to restart one of the decommissioned reactors and grant financial subsidies to any private companies willing to construct new power plants. In 2023, the government accordingly appointed an inquiry to investigate the construction of new nuclear power plants in Sweden (Regeringen, 2023a). By the end of 2023, the government also presented its Climate Action Plan, in compliance with the Climate Act, in which it claimed that "… in Sweden nuclear power is the single most important measure to reduce our emission nationally …" (Regeringen, 2023b, p. 10).

3.2. The divided support for wind power

The expansion of wind power has also become an issue characterized by growing political tension. Wind power has developed rapidly in Sweden, and in between 2001 and 2021, the installed effect increased by almost ten times (Swedish Energy Agency, 2021). The expansion of wind power has been widely supported in society, yet in recent years, attitudes has become increasingly polarized (Jönsson, 2022).

Just like the debate on nuclear power, political opinions are divided between conservative parties, which express anti-wind power rhetoric, and the more supportive center-liberal, green, and left-leaning parties (Axelsson et al., 2023). At the local level, wind power has become a particularly divisive issue. The Environmental Act and the municipal planning monopoly grant municipalities the power to decline construction permits for wind power sites, colloquially referred to as the "municipal veto." The number of municipal rejections has increased in recent years, and in 2021, around 80 percent of all construction applications were declined (Westander, 2022). A key reason for these rejections is the lack of formal compensation for people living near wind power sites (Lindvall, 2023). As a result, the government commissioned a public inquiry in 2021 to present proposals for a more predictable municipal siting process (SOU, 2022:53), and in 2022, another inquiry was appointed to introduce local incentives for wind power development (SOU, 2023:18). The new conservative government, elected in 2022, expressed a negative stance on wind power during the election campaign and has withdrawn public subsidies for the connection cost of offshore wind power to the power grid. In its Climate Action Plan, the government eased its negative policy stance towards wind power, although it did not present any proposals for facilitating wind power development or overcoming acceptance barriers (Regeringen, 2023b).

4. Research hypotheses, methodology and data material

4.1. Hypothesizing determinants for acceptance of energy investment

The aim of this article is to examine potential determinants of social attitudes to wind and nuclear energy, based on the results of an exploratory survey. Previous studies have demonstrated that environmental concern is a key determinant for both wind and nuclear power, but also that political attitudes and values, such as ideology,

worldviews, and personal identity, influence attitudes to these energy options (Karlstrøm and Ryghaug 2014; Clulow et al., 2021; Pampel, 2011; Whitfield et al., 2009). Based on these previous findings, we assume that individuals with right-leaning attitudes and low climate concern, express higher support to nuclear energy, while left-leaning individuals with high climate concern, express higher support for wind power. We also assume that worldviews, that correspond with traditional, authoritarian, and nationalistic (TAN), or green, alternative, and libertarian (GAL) values (Hooghe et al., 2002), predict energy preferences, as well as political and institutional trust. To explore these correlations, we will test the following hypothesis.

H1. Value -based orientations influence peoples' propensity to support nuclear and wind power.

In a situation with a high level of political polarization between different energy policies, it is likely that energy preferences are determined by the attitudes toward one technology. Such effects can indicate that attitudes are determined by political cues (Latré et al., 2019) or politically motivated reasoning (Kahan, 2013), rather than environmental concerns, risk perceptions, economic benefits, energy security or other factual considerations. There could, thus, be a correlation between support for nuclear power and negative attitudes to wind power, while the correlation is reversed for support for wind power. Such a relationship could indicate that polarization is a factor that can explain attitudes. To explore this, we will test the following hypothesis.

H2. The propensity to support nuclear power is determined by negative attitudes toward wind power.

Another factor that has been shown to strongly influence support for wind power is the location of the sitings (Batel et al., 2013; Carley et al., 2020). The closer a wind farm is constructed to where people live, the less likely they are to accept it. This proximity effect has not previously been shown to strongly affect people's attitudes to nuclear power, yet it is uncertain whether people hold supportive attitudes prior to the construction process. It is, moreover, not clear how value-based orientations interact with, or alleviate, the proximity effect, which could indicate how strongly attitudes are underpinned by cultural values. To explore these associations, we test the following three hypotheses.

H3. The proximity of a wind power project predicts attitudes toward wind power development.

H4. The proximity of a nuclear power project predicts attitudes toward nuclear power development.

H5. Value-based orientations mediate the proximity effect on attitudes toward nuclear and wind power.

4.2. Survey design and data

The study is based on data from a large-scale public opinion survey on attitudes to climate and energy policies in Sweden. The survey was conducted within the research project Fairtrans by the survey company Skop. It included questions on both climate and energy policies, and a subset of the questions was analyzed for this article. The sample was drawn from the population register and made as a non-proportional stratified sample in three age categories and divisions between different types of municipality, in order to attain a proportional representation. It was sent out in postal format, with possibilities for digital responses, to 33 432 respondents aged 18 to 84 in between 25 April and 23 August, 2023. A post-stratification methodology was applied to make the sample representative of the total population. After weighting the dataset on a number of core demographics to make it representative of the Swedish population, a total of 5280 respondents were included in the dataset. More information about the data collection and stratification methodology is provided in the supplementary material to this article.

To capture both general attitudes and proximity effects, participants were asked to respond to three statements regarding each energy technology. The statements were phrased; To speed up the transformation to a fossil free society, I support further construction of wind/nuclear power in Sweden. Thereafter, the respondents were asked if they support (1) construction of wind/nuclear power in their municipality and (2) construction of wind/nuclear power within 5 km from their home. Five kilometers was estimated to be considered as intrusive, realistic, and comparable between the two energy options. For each statement, the respondents were asked to indicate their level of agreement on a seven-point scale. The scale was anchored at 1 = *Completely disagree*,[1] 4 = *Neither/nor*, and 7 = *Completely agree*.

We thereafter designed questions on value-based orientations, aspiring to both capture positions on the left/right political spectrum and social ideals and worldviews. On the later issue, we used a scale inspired by the GAL (Green/Alternative/Libertarian) vs. TAN (Traditional/Authoritarian/Nationalistic) framework (Hooghe et al., 2002). GAL/TAN values are typically assessed using a set of value judgments regarding aspects such as multiculturalism, international solidarity, gender equality on one end, and conservatism and national community on the other (Hobolt et al., 2021). GAL values were assessed on the responses to the following two questions: "*Society is enriched by open borders, global cooperation and international solidarity*" and "*It is important to promote equality between men and women*". TAN was assessed on the following two questions: "*In a secure society, immigration is low and the majority of the inhabitants share the same culture*" and "*In a secure society, law and order is maintained by a strong police force and a judiciary that can sentence cognizable punishments for criminals*". GAL is the weighted responses of the two first questions, and the opposite for TAN.

Two questions aimed to measure trust were created, asking respondents to assess if they believed politicians to most often act in the public interests (political trust) and if public authorities (governmental trust) most often act competently, efficiently and in accordance with the rule of law. The same response scale as described above was used.

To capture environmental concern, a subset of Schultz's scale (Wesley schultz, 2001) was used. Respondents were asked to indicate their level of concern about environmental problems and their consequences to (1) myself, (2) all people, (3) my lifestyle, (4) people close to me, (5) animals and plants, (6) future generations, and (7) children, on a seven-point scale. The scale was anchored at 1 = *Not at all concerned* and 7 = *Very concerned*. The responses were collated into an index assessing environmental concern.

Other individual characteristics gathered in the survey regarded income, education, place of living (rural/urban), and gender.

By conducting a series of OLS regression models, we examined if preferences for different energy technologies are influenced by value-based orientations and sociodemographic factors, testing aspects of proximity, and also using baseline attitudes to the two energy options as independent variables.

5. Results

5.1. Attitudes to wind and nuclear power

Attitudes to wind and nuclear power is evidently polarized in Sweden. Fig. 1 reports the distribution across the different response options on the seven-point scale. A general tendency, is that antagonistic views (i.e., responding 1 on a seven-point scale) of both wind and nuclear power is relatively frequent; systematically more frequent than negative, but less antagonistic views (i.e., responding 2 or 3 on a seven-point scale). Similarly, a very positive view (i.e., 7) of more wind and more nuclear power is more frequent than a positive but less extreme view (i.e., 5–6). Taken together, the response distribution suggests a tangible

[1] There is no comparable term for *disagree* in Swedish so the phrase *Instämmer inte alls* was used, which literally can be translated to *Do not agree at all*.

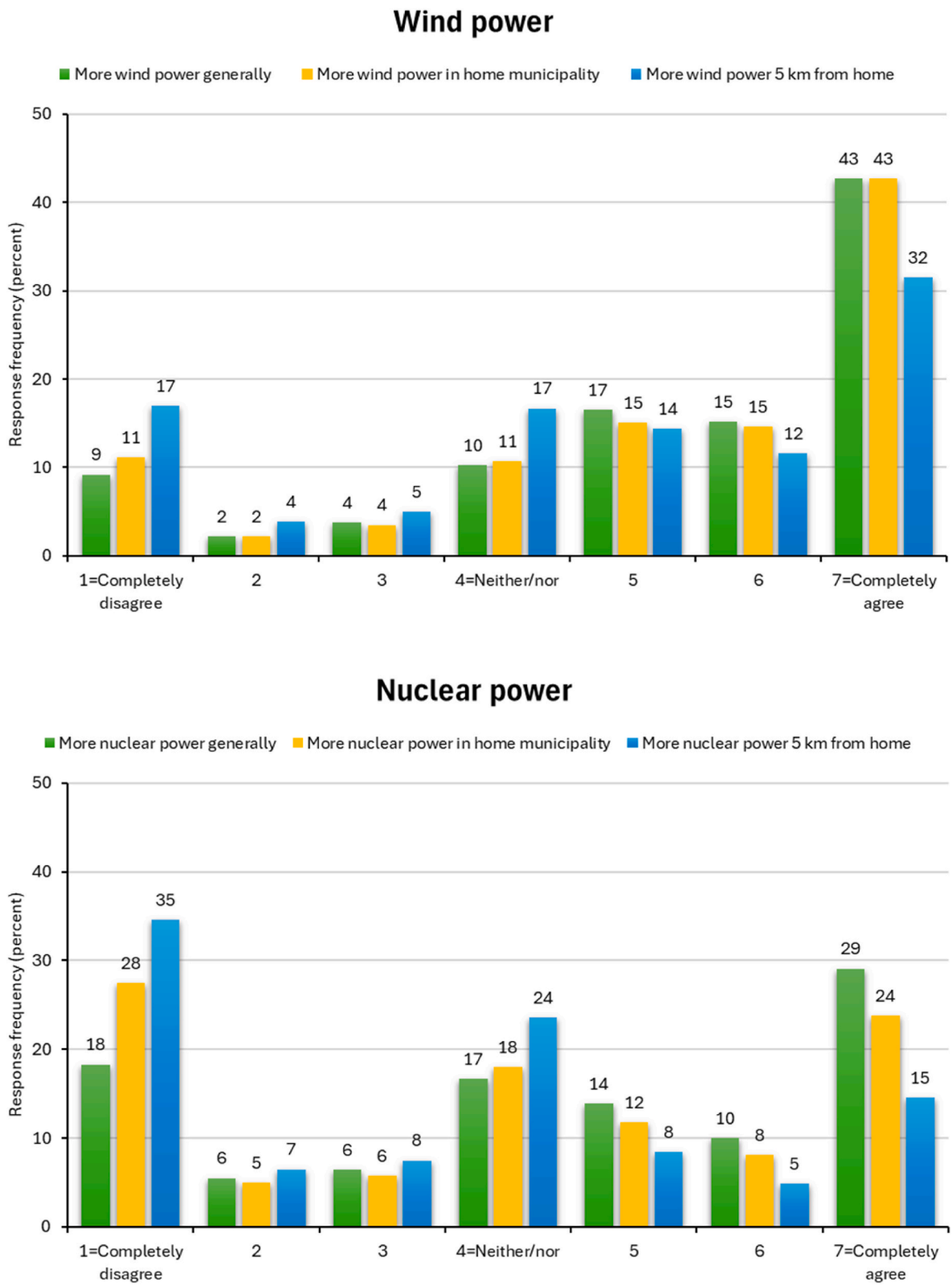

Fig. 1. Response distribution for attitudes toward wind and nuclear power.

polarization in the respondents' views of wind and nuclear power.

Fig. 2 reports average attitudes to nuclear and wind power. On average, participants were more positive to the development of more wind than nuclear power, $t(5231) = 19.04$, $p < .001$, Cohen's d = .26. As expected, positive attitudes to both nuclear and wind power declined with proximity. The closer the development of new power, the less positive attitudes.

Table 1 reports correlations between the main variables. A number of things are worth pointing out. Attitudes toward more nuclear power and attitudes toward more wind power were negatively correlated. Thus, a greater acceptance for more nuclear power was associated with a lesser acceptance for more wind power, and vice versa. A higher

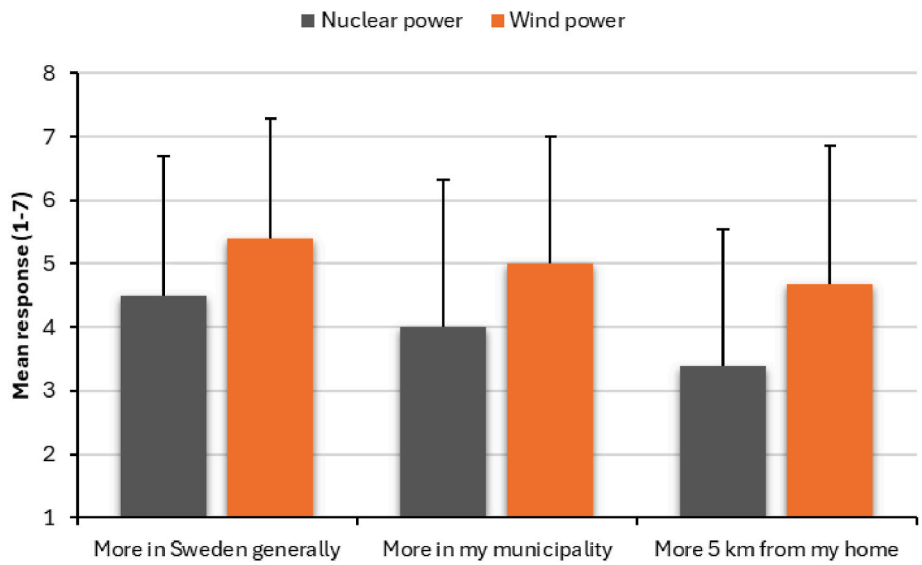

Fig. 2. Average attitudes toward more nuclear and wind power.
Note. Error bars represent 1 *SD*. Estimates were made on a 7-point scale where 4 represented a neutral standpoint. Values above 4 represented a positive attitude to more nuclear/wind power and values below represented a negative.

Table 1
Correlation coefficients for the relationships between main variables.

Variable	1	2	3	4	5	6	7	8	9	10
1. More NP generally	–									
2. More WP generally	−.37*	–								
3. More NP 5 km from home	.70*	−.31*	–							
4. More WP 5 km from home	−.31*	.72*	−.15*	–						
5. Environmental concern	−.35*	.46*	−.32*	.35*	–					
6. Political ideology	−.49*	.36*	−.34*	.32*	.36*	–				
7. TAN	.51*	−.37*	.36*	−.32*	−.33*	−.52*	–			
8. GAL	−.36*	.47*	−.28*	.38*	.45*	.42*	−.52*	–		
9. Trust in politicians	−.05*	.24*	−.07*	.17*	.17*	.02	−.10*	.23*	–	
10. Trust in governments	−.18*	.35*	−.17*	.26*	.25*	.17*	−.25*	.36*	.63*	–
11. Salary	.07*	−.008	−.07*	−.002	−.008	−.08*	.03	−.02	.02	.009

Note: NP = Nuclear power; WP = Wind power.

environmental concern (M = 5.04, SD = 1.36) was associated with a lower acceptance of nuclear power, but a higher acceptance of wind power. These relationships are presumably driven by individual differences in political ideology (M = 4.20, SD = 1.63), as a greater left-wing political lineage was also associated with higher environmental concern, a more negative attitude to nuclear power and a more positive attitude to wind power. Similarly, higher TAN (M = 4.44, SD = 1.65) was associated with a more positive attitude to nuclear power but a more negative attitude to wind power, whereas GAL (M = 5.65, SD = 1.21) followed the opposite pattern.

Besides the influence of such values, higher trust in politicians (M = 3.72, SD 1.55) and in governments (M = 4.33, SD = 1.57) were weakly and negatively related to attitudes toward nuclear power, but more strongly and positively related to attitudes toward more wind power. We can also conclude that self-reported salary (M = 45. 85 thousand SEK per month, SD = 68.26) had no tangible correlations with the other variables. Small but significant correlations were found between salary and acceptance of more nuclear power and political ideology. A higher salary was associated with a greater acceptance of more nuclear power and a preference for right-wing political ideology.

There were no statistically significant differences between people living in rural versus urban areas, with regard to their attitudes to more nuclear power. However, there were clear gender differences, showing that men were consistently more supportive of nuclear power than women. Men were positive to more nuclear power generally (M = 5.06, SD = 2.12) whereas women were not (M = 3.91, SD = 2.14), $t(5176) = 19.38$, $p < .001$, Cohen's d = .54. Men were also, on average, positive to more nuclear power in their home municipality (M = 4.66, SD = 2.26) whereas women were not (M = 3.35, SD = 2.16), $t(5173) = 21.31$, $p < .001$, Cohen's d = .59. In turn, men were, on average, slightly negative to more nuclear power within 5 km from their own home (M = 3.92, SD = 2.23) whereas women were clearly negative (M = 2.81, SD = 1.92), $t(5172) = 19.04$, $p < .001$, Cohen's d = .53. These gender differences likely emerged because of a difference between genders regarding environmental concern, with women (M = 5.38, SD = 1.17) reporting higher concern than men (M = 4.72, SD = 1.45), $t(4875) = 17.33$, $p < .001$, Cohen's d = .50.

There were also small, but statistically significant differences between respondents with higher education (N = 3054) and respondents without higher education (N = 2193). People with higher education were positive to more nuclear power generally (M = 4.16, SD = 2.26), but less so than people without higher education (M = 4.73, SD = 2.13), $t(5245) = 9.31$, $p < .001$, Cohen's d = .26. People with higher education were, on average, negative to more nuclear power in their home municipality (M = 3.68, SD = 2.32), whereas people without higher education were positive (M = 4.25, SD = 2.27), $t(5242) = 8.91$, $p < .001$, Cohen's d = .25. In turn, both people with higher education (M = 3.09, SD = 2.14) and people without higher education (M = 3.59, SD = 2.15) were, on average, negative to nuclear power within 5 km from their own home; people with higher education more so, $t(5241) = 8.28$, $p < .001$,

Cohen's d = .23.

To further explore the polarization of attitudes, we conducted an additional analysis of the characteristics of respondents with extreme positions. Specifically, we looked at respondents who were extremely positive to more nuclear power generally (i.e., answered 7 on the seven-point scale) and at the same time extremely negative to wind power generally (i.e., answered 1 on the seven-point scale) — henceforth called pro-nuclear anti-wind group (N = 379); and we also looked at the group of respondents with attitudes in the opposite direction — henceforth called pro-wind anti-nuclear group (N = 688). These two groups amount to a total of around 20 percent of the total sample and their characteristics are reported in Table 2. We can conclude that the pro-wind anti-nuclear group was characterized by higher environmental concern, a left-wing political ideology, higher trust in governments, higher trust in politicians, a higher GAL, much lower TAN, and a slightly lower salary than their counterpart. The analysis reveals accordingly that worldviews, ideology, and environmental concern are key aspects determining extreme positions.

5.2. The effect of proximity

A diff scores analysis was conducted to test if values moderate the proximity effect of nuclear power. In the first step, the difference between attitudes to more nuclear power generally and attitudes to more nuclear power five km from peoples' homes was calculated for each participant respectively. This diff scores variable was then entered into a regression model as a dependent variable. Political ideology, environmental concern, TAN, GAL, trust in politicians, trust in governments, and self-reported salary were selected as independent variables. The results are reported in Table 3.

The analysis of attitudes to nuclear power projects close to peoples place of residence reveals that political ideology, TAN and, to lesser degree also environmental concern, explains unique variance of the proximity effect. Interestingly, a higher TAN, a higher environmental concern, and a more right-wing political lineage were associated with a larger proximity effect. That is, a larger difference between acceptance of nuclear power generally and nuclear power near one's own home. This effect emerges because people with low TAN and left-wing political ideology were consistently negative to nuclear power regardless of its localization. For them, the acceptance of nuclear power were similarly low across localization and the proximity effect was small in magnitude. In contrast, people with high TAN and a right-wing political ideology were relatively positive to nuclear power, especially so when not in their own proximity. Hence, the proximity effect is thus larger among this group of participants.

Table 2
Means and standard deviations (SDs) across attitude-related factors for respondents in two extremely polarized groups with opposed attitudes.

	Pro-nuclear anti-wind		Pro-wind anti-nuclear	
	Mean	SD	Mean	SD
Environmental concern	3.21	1.66	5.77	.91
Political ideology	2.66	1.42	5.38	1.28
Trust in governments	2.83	1.71	4.72	1.54
Trust in politicians	2.53	1.62	3.70	1.54
GAL	4.26	1.46	5.65	1.21
TAN	6.19	1.19	3.22	1.57
Monthly income	49.93	83.98	43.10	68.23

Note: GAL = Green-alternative-libertarian; TAN = Traditional-authoritarian-nationalist. A political ideology above 4 represents a left-wing ideology, a value below 4 represents a right-wing ideology.

Table 3
Results from a multiple regression model with the proximity effect (difference between acceptance for nuclear power generally and nuclear power 5 km from home) as dependent variable.

Predictor/control variable	Beta	*t*	*p*
Environmental concern	.05	2.83*	.005
Political ideology	−.14	7.62*	<.001
Trust in governments	.003	.17	.862
Trust in politicians	.04	1.88	.060
Green-alternative-libertarian (GAL)	−.008	.43	.671
Traditional-authoritarian-nationalist (TAN)	.15	7.74*	<.001
Monthly income	−.01	.68	.494

Note: * flags statistically significant values.

A similar diff scores analysis was conducted to test if values moderate the proximity effect of wind power. The results are reported in Table 4. In this case, environmental concern was the strongest predictor. The only other predictor that reached significance was trust in governments. Again, we note a positive relationship between the predictor variable and the proximity effect. A higher environmental concern was associated with a greater proximity effect, although the correlation was not as strong as TAN values and political ideology were for nuclear power. This is driven by the same mechanism as the relationships for nuclear power acceptance. A higher environmental concern is associated with a greater acceptance of wind power, especially when the wind power is built somewhere that is not within the proximity of one's own home. People with a low environmental concern were, in turn, generally negative to wind power regardless of its localization, thus leading to a smaller proximity effect.

6. Discussion

Our study reveals that individual values notably forecast attitudes towards both wind or nuclear power, aligning with our initial hypothesis. Consistent with prior studies, it is evident that environmental concern significantly shapes the views on wind power (Arndt, 2023; Azarova et al., 2019; Perlaviciute and Steg, 2015), and resistance to nuclear power (Whitfield et al., 2009; de Groot et al., 2013). Wind power can impact on biodiversity and the local environment (Adeyeye et al., 2020), still in our study environmental concern was strongly associated with wind power support, even though we captured environmental attitudes with the usage of a broad measurement, not only related to for climate change. It can be assumed that a great number of people associate nuclear energy with ecological and human risks, as suggested by Whitfield et al. (2009), and this can partly explain why environmentally concerned individuals are skeptical to nuclear energy. The assumption that nuclear energy has become popular as a low-carbon alterative in the energy transition, as suggested by Bohdanowicz et al. (2023), is not supported by this study. To overcome this type of resistance, policymakers may consider awareness campaigns, addressing concerns regarding safety of nuclear energy, while emphasizing its potential environmental benefits.

Table 4
Results from a multiple regression model with the proximity effect (difference between acceptance for wind power generally and wind power 5 km from home) as dependent variable.

Predictor/control variable	Beta	*t*	*p*
Environmental concern	.07	4.24*	<.001
Political ideology	−.02	1.26	.207
Trust in governments	.04	2.16*	.031
Trust in politicians	.03	1.58	.114
Green-alternative-libertarian (GAL)	.002	.09	.932
Traditional-authoritarian-nationalist (TAN)	.02	.95	.341
Monthly income	−.01	.97	.333

Note: * flags statistically significant values.

The strongest determinants for support or resistance to wind and nuclear energy are, however, worldviews and ideological orientations. Nuclear energy is strongly correlated with right leaning political orientation, while individuals with left leaning political views, are more supportive of wind power. It is unclear if the respondents had full understanding of the economic costs or benefits of the two energy options. Personal beliefs and values may, on the other hand, influence public receptiveness of information about different energy alternatives.

Attitudes are also likely to be subject to political polarization. A rather remarkable finding in this regard was the exceptionally high correlation between social ideals and cultural values and different energy options. Individuals who embrace openness, multiethnic tolerance and gender equality prefer wind power, while those expressing authoritarian and patriotic worldviews, favor nuclear energy. It can thus be concluded that energy politics in Sweden is divided along the so-called GAL/TAN scale (Hooghe et al., 2002).

Interactions between opinions about energy preferences and worldviews, e.g. individualistic, egalitarian or hierarchical worldviews, have been found in previous studies (Van De Graaff, 2016; Perlaviciute and Steg, 2015). Explaining these associations, Groh and Möllendorff (2020) claim that proponents of nuclear energy tend to take human dominance over nature for granted, and therefore have less anxiety of the human and ecological consequences of potential accidents. Upon this assumption, it is also reasonable to believe that individuals who are not endorsing principles of gender equality nor displaying strong solidarity with other people, take a more dominant position in society and in relation to nature, in accordance with the theory of Social Dominance Orientation (Pratto et al., 1994). Previous research has shown that groups that express group-based dominance tend to justify or promote social and racial inequalities, hierarchies and discrimination (Jost and Thompson, 2000), while there are also correlations between social dominance and ignorance of environmental degradation (Milfont and Sibley, 2014) and climate denial (Jylhä and Akrami, 2015).

High TAN and low GAL scores may thus reflect a similar predisposition towards social dominance orientation. Notably, individuals with such orientations exhibited heightened endorsement for nuclear energy and resistance towards wind power. Consequently, it can be inferred that values aligning with social hierarchy correlate with a belief in human dominance over nature. This inclination may lead individuals to downplay the potential risks associated with nuclear energy and overemphasize negative aspects of wind power. In an inverted manner, individuals who display solidarity for other people and advocate for gender equality, are instead likely to incorporate concerns for the environment, and thus being more skeptical to nuclear power and more supportive for wind energy (Whitfield et al., 2009).

Another plausible interpretation of the correlations between value expressions and energy opinions could be attributed to the political polarization inherent in Swedish energy politics. It is probable that individuals whose political affiliations and cultural values align with the extreme ends on the ideological spectrum are swayed by political debates, and that they adjust their opinions accordingly. Political parties have shown to have substantial influence over public opinion formation, as voters are often guided by political cues (Goren et al., 2009), and this may explain the polarization of energy preferences.

With increasing ideological polarization, political views can moreover be transformed into identities, shaping our wider interpretations of reality and becoming aspects motivating our reasoning about various political objects (Cohen, 2003), including energy technologies. The psychological urge of belonging to a social group can in this sense contribute to furnish our worldviews, making us unconsciously conform our views in line with the opinions expressed by communities that share our identity and political affiliation. Facts that are in conflict with group membership can therefore trigger a form of unconscious identity self-defense, which motivate individuals to resist empirical facts (Kahan, 2013). There is robust evidence for the influence of motivated reasoning for climate policy formation (Druckman and McGrath, 2019), and it can, thus, be expected that the same phenomena emerge in energy politics, particularly in Sweden where the debate has been characterized by deep polarization. To measure trends of group polarization and actual prevalence of motivated reasoning, we would need longitudinal data to capture changes of attitudes over time, and for this reason we cannot draw any firm conclusions on these matters. We did find support, however, for our second hypothesis, noticing that negative wind power attitudes determine positive attitudes to nuclear power, and vice versa.

Another remarkable finding was the strong correlation between governmental trust and energy opinions. High trusting individuals are supportive of wind power, while individuals with low trust prefer nuclear power. At face value this correlation appears contradictory, given that it can be presumed that people who lack trust in public institutions would not support an energy technology that require expert involvement and an institutional infrastructure to operate safely. This result also contradicts previous findings regarding associations between trust and nuclear power support (Siegrist et al., 2000; Van De Graaff, 2016). A potential interpretation of this correlation is that governmental trust is manifesting anti-establishment attitudes. While nuclear energy once was seen as a centralized technology associated with a "technoscientific-industrial-military elite" (Whitfield et al., 2009), it has now become an energy option supported by groups distrusting elites, expressing values that correspond with the rhetoric pursued by right-wing parties. Wind power, on the other hand, which is currently advocated by many energy experts on the basis of its economic and environmental merits, is seen as the energy technology of a new type of elite, at least from the perspective of individuals who distrust scientific experts, express a low level of environmental concern, and identify as right-wing politically. This view of a "wind power elite" might thus be deliberately created, producing mental distance to technologies that are associated with the opposite, left-wing political side. In this sense, a conceptual shift has occurred, affecting the perceptions of these technologies, which is caused by ideological, normative, and political transformation of society, rather than changes of these technologies as such. This finding presents another obstacle for socially acceptable energy policy solutions. If political parties negotiate agreements on energy policy, as suggested above, they may inadvertently pander to such anti-establishment sentiments.

Moreover, this study suggests that general attitudes to various energy options, and the factors determining them, differ from attitudes to energy investments in the vicinity of people's place of living. While the majority of respondents express a willingness to embrace wind power installations in their municipality or near their homes, the study also endorses the third and fourth hypotheses regarding the proximity effect. There was particularly strong reluctance towards construction of nuclear power facilities near respondents' residences. Even though few previous studies have explored this effect, the discovery was not surprising. This type of acceptance barriers would be relevant to consider in case small modular reactors will be successfully developed and deployed in the future.

An unanticipated finding was the variance in the proximity effect among groups with different value orientations. As shown by our analysis, individuals with high TAN and a right-wing political ideology became relatively more negative to nuclear power when it was suggested that power plants were to be built near their places of living, and a similar effect could be noticed for the influence of environmental concern on wind power attitudes. In regards of our final hypothesis, we can thus conclude that proximity moderates value-based orientations, rather than the other way around. To put it differently, strong attitudes do not protect against the negative effects of proximity. Proximity is given higher weight in the acceptance judgment, leading to the perhaps counterintuitive conclusion, that people with more favorable views of a specific energy source are more susceptible to the negative effects of proximity, because their judgments of acceptance change to a greater degree when proximity increases. In turn, people who already have a negative view of a specific energy source are not affected by proximity as

they remain sceptic regardless of proximity. A possible interpretation of this finding is that politically motivated reasoning becomes less relevant when people are asked to judge energy projects in the proximity of their homes and, thus, affecting them personally. If this is correct, these individuals may perhaps also consider other types of policy measures that affect them directly, by for instance providing financial compensation and community benefits. Such measures have shown to influence social acceptance for wind power (Liebe et al., 2017; Knauf, 2022), yet their effect on nuclear energy acceptance is unknown. These effects need to be further explored, preferably by studying actual lived experiences of individuals in residences nearby planned and constructed energy production facilities.

7. Conclusions and policy recommendations

Our exploration of attitudes to different energy options in Sweden demonstrates that while both wind and nuclear energy enjoy fairly strong public support, opinions are distinctly divided. As many as 20 percent of the total sample was strongly in favor of one energy option, while at the same time being strongly against the other, and thus expressing a seemingly uncompromised stance. While opinions are polarized, public views are first and foremost determined by worldviews and ideological orientations. Policymakers aiming to further decarbonize the energy system must thus be prepared to address opposition and navigate potential tensions between different segments of society, regardless of which low-carbon energy option they choose.

These findings highlight how important it is to recognize the socio-psychological dimensions of different energy options and the social and normative entanglements of energy systems. Addressing resistance that is deeply anchored in values and worldviews is difficult, however, particularly as political polarization and the focus on identify and value-related conflicts has become commonplace in the political debates in many western democracies. There is a risk that political cues further transform energy policies into a matter of identity and values, and generate impasses in the energy transitions. Policymakers that are serious in their ambitions to speed up the process of decarbonization should therefore make attempts to overcome the partisan divide. This could be done by negotiating interparty agreements on energy policies, such as the energy agreement in Sweden that was dissolved in 2019.

CRediT authorship contribution statement

Daniel Lindvall: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Patrik Sörqvist:** Writing – review & editing, Formal analysis. **Sofie Lindeberg:** Writing – review & editing. **Stephan Barthel:** Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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